

AI is actually net positive for the environment

Posted by u/Reynvald • 0 points • 50 comments

Take all calculations and sources here with a grain of salt for both sides of the arguments, as such things are generally hard to quantify. I also would be happy to get corrected if I made mistakes or misrepresented some data. And yes, I used various AI tools for research, but manually checked every source that I put in here.

Usual talking points about AI, harming the environment, is:

- Energy consumption
- Carbon footprint and GHG in general
- Water scarcity

1. Energy consumption

As of 2024, [Data centers accounted for about 1.5% of global electricity consumption, with AI accounted for 15% of total data centre energy demand accordingly.](<https://www.scientificamerican.com/article/ai-will-drive-doubling-of-data-center-energy-demand-by-2030/>) Therefore we can say that AI itself is using around 0.225% of global energy reserves.

[Predicted share of energy usage for data centers by 2030 is between 5 and 20%.]

(<https://www.scientificamerican.com/article/ai-will-drive-doubling-of-data-center-energy-demand-by-2030/>) Considering that AI is still on its growth and can take over up to 50% of all data center's resources, in 2030 it can be responsible for 2.5 up to 10% of all energy consumption (20 up to 90 times more, than of now) which is quite radical prediction.

Nevertheless, as of right now, [ML-related technologies is able to provide 15% improvement in grid efficiency and 10–20% increase in battery storage efficiency](<https://www.mdpi.com/2073-4433/15/10/1250>) and [20–30% relative efficiency gains in cell and module R&D.]

(<https://journals.sagepub.com/doi/full/10.1177/0958305X241256293>)

Same magnitude of efficiency gains is also the case for all clean and non-clean energy sources, by forecasting the weather and autoadjusting solar panels, micromanaging power grids and plants, predicting deposits of fossil energy sources and so on.

Safe to say, that estimated energy gain overall will equal to or most likely surpass even the most pessimistic prognosis of 10% energy consumption from AI alone by 2030.

2. Carbon footprint and GHG in general

According to [ICEF report from November 2024](https://www.icef.go.jp/wp-content/themes/icef/pdf/2024/roadmap/15_ICEF2.0%20GHG%20Emissions%20from%20AI_stand%20alone.pdf), (This link will download PDF file!) AI's total GHG emissions are estimated at 100–300 million tonnes CO₂, or roughly 0.2–0.6% of global emissions. With that, operational emissions are around 0.05% while manufacturing servers, chips, facilities, model trainings and life-cycle impacts make up the remainder.

At the same time AI can [reduce global GHG emissions by 5–10% by 2030] (<https://thesustainableagency.com/blog/environmental-impact-of-generative-ai/>), via optimized grids, predictive maintenance, and smart agriculture and, additionally, [cuts of up to 5.3 gigatons CO₂ (another 5–10% of current emissions)](<https://www.sciencedirect.com/science/article/pii/S277242712200122X>) - through applications in transport, buildings, and supply chains.

[One specific research](<https://www.nature.com/articles/s41598-025-97110-3>) (from month ago) from China indicates, that correlation between % of AI adoption and % of reducing carbon footprint (1% and 0.0395% accordingly) is quite sustainable and universal across the industries.

3. Water scarcity

There is not much fresh unbiased data and peer-reviewed papers on AI water consumption. [Apparently in US AI is responsible for 0.5–0.7% of total annual water withdrawal](<https://oecd.ai/en/wonk/how-much-water-does-ai-consume>). If source took a data of water consumptions by data centers in general (it most likely the case), then actual numbers will be a 15% of 0.5–0.7%, which is 0.075–0.105% accordingly.

Considering that most of the world AI infrastructure is located in US and China, safe to say, that for the rest of the world this percentages is significantly smaller.

The real concern, however, is the water pollution (which is still extremely small, compared to the heavy and construction industries) and separate cases of mismanagement from the corporations. [Quote: "Google's planned data centre in Uruguay, which recently suffered its worst drought in 74 years, would require 7.6 million litres per day, sparking widespread protest."](https://sdgs.un.org/sites/default/files/2024-05/Gupta%2C%20et%20al._AIs%20excessive%20water%20consumption.pdf) (This link will download PDF file!)

Now to a good news:

[AI irrigation can reduce water usage by 30–50% while increasing yields by 20–30% (which is 5–8% savings of global agricultural withdrawals if deployed worldwide).] (<https://www.sciencedirect.com/science/article/pii/S2772375525002151>)

[AI acoustic and pressure-based leak detection is already working and have 80–97% accuracy, cutting non-revenue water losses by 20–40%. Given that networks lose ~30% of supply globally (the most distant and arid places usually suffer the most), AI is saving 6–12% of treated water.] (https://www.waterrf.org/serve-file/2024_IWS-Challenge-Solution_Case-Western-CWD.pdf) (This link will download PDF file!)

Same goes for demand forecasting, pump optimization, water quality assessment and many other projects, totaling up to 12% of the saved fresh water worldwide (if implemented worldwide as well). Some of this solutions is already implemented and working, although mostly in the most water hungry areas, like parts of Africa, China and India.

There is crucial to point out, that most of the water scarcity-related suffering is occurring far from data centers and their water sources. And this problem is a logistical one (how to transport the water to the arid areas), not the problem of sheer amount of fresh water world supplies.

Fun facts, regarding the general misconception that AI consume literally bottles of water per query:

.1. [The amount of water, that ChatGPT needs to consume is around 500ml of water *for 10-50 queries*

This means that each query is about $500/30=17\text{ml}$.]

(<https://miljamoss.neocities.org/Articles/LLMWaterAndEnergyUse>)

- The amount of water required to produce an 8oz steak is 3,217,000 ml. So you would need to make 189,000 queries to equal the water cost of a steak dinner.

- Average shower uses about 8000ml of water per minute. So you'd have to make 470 queries to use the amount of water you spend if you're in the shower for one extra minute.

- Finally, flushing the toilet uses 6000ml. So if you pee one extra time per day that's about 350 queries.

.2. [Humans themselves is far more environmentally impactful, compared to AI, when performing the same tasks. Hundreds times so, even.](<https://www.nature.com/articles/s41598-024-54271-x>)

I want to highlight, that AI still have an impact on environment and it's a right thing to strife for reducing the environmental impact in any area. But I believe that misinformation, toxicity and alarmism eventually will harm the both sides of this debates.

Comments

u/lapetus_Industrial • 2 points • 2025-05-12 20:30 UTC

AI has also made advancements in more efficient matrix multiplications, which _all_ server farms use for just about everything. Might not seem too impressive but the savings add up to billions saved in energy usage overall - as in in computation that already was going to happen with or without AI.

u/Reynvald • 2 points • 2025-05-12 21:48 UTC

No, it's actually quite impressive. Any improvement on such base scale will always results in tremendous significant of the system as a whole. I will read about it later, thanks for the tip.

You made quite a mistake in your first 2 points, didn't feel like reading the rest.

You give stats that are about "normal" AI when it's about the positive impacts. The cost, though, comes mostly from generative AI, especially creating picture, which has nothing to do with making energy grids or transportation more efficient.

So we could get most if not all the benefits of AI by using specialized models, and probably wouldn't lose much by just shutting down chatgpt and the likes. Not that it's what I'm arguing for, just saying. I just wish we'd be smarter about the whole thing, but alas, this will soon be AI's job.

u/Reynvald • 2 points • 2025-05-11 17:58 UTC

I would argue with it on several occasions.

- .1. You can't really create a technology and then limit it for narrow window of cases. It might be good for some things, but it just almost never works. So we have to take it as a given. Besides, each query is still have a complete negligible impact, compared with everyday things that we taking for granted.
- .2. Using LLM for everyday life (considered that it evidently save people's time), might also have it's environmental advantages. According to Nature journal (last link in my post) LLM much lest wasteful then humans at the respective tasks. Like hundreds times less wasteful.
- .3. There are many strong studies, that shows measurable proofs of increasing quality of education when LLM is used appropriately. It's 100% the same "normal" AI you are referring to. And more educated people are — better results for humanity in every endeavors.
- .4. The usage of "Normal AI" is precisely what sponsoring development and implementation use-case specific AI, attributed to positive environmental impact.

u/Lost-Basil5797 • 0 points • 2025-05-11 18:58 UTC

Good points! And I'm 100% convinced that when used right, it's a great tool for many things. I'm more worried about the "used right" and how humanity seem to often fail at that very step. Internet itself, if used right, has a huge potential for transformation. And yet here we are, with the biggest platforms being basically made to make people addict to scrolling infinitely, the public discourse being worse than ever as division is rampant and shit-flinging based arguments so common. List could go on. I'm still a firm believer that it could be used way better than that, and hopefully will be, but as is, I'd be hard pressed to find more positives than negatives to the state of internet.

We'll see, though!

u/Reynvald • 3 points • 2025-05-11 19:07 UTC

Yea, I actually can relate. I also see many potential risks, current and future ones. Reduction of critical thinking and parasocial relationships (especially with this clearly marketing decisions to make LLMs sycophants), bad actors, massive job losses and another "lost generation" as a result. Not mentioning AI takeover (although not much people sees it as a real possibility). But yeah, we'll see :D

u/Lost-Basil5797 • 1 points • 2025-05-11 19:39 UTC

>Reduction of critical thinking and parasocial relationships (especially with this clearly marketing decisions to make LLMs sycophants)

Honestly I really fear for that one, your parenthesis very much included. These last years I've started to consider that culture might be more important than the political system of a given population. If, for exemple, your culture promotes cultivating your own wealth at the expanse of anyone and anything, it doesn't really matter if you live in a capitalistic or communist country, same with democracy or royalty. It does matter, of course, but I could see any of these work similarly well

if, say, everyone had to heart to play well with people and put "the greater good" as a priority.

So back to AI, the tool is great, probably one of our greatestests like fire, the wheel and printing. But when I look at the rising cultures in the modern world, the ones birthing said tool...

Yikes.

u/Reynvald • 2 points • 2025-05-11 19:58 UTC

For my 30 years I've been a monarchist, socialist, liberal, libertarian and anarchist, to eventually become a political realist. Play with the cards you got, try to do the right things, even if the Big Brother above you is going nuts. Hell, repaint the benches around your building if you unable to do more at the moment. I guess such position is to be expected from a Russian citizen :D

As for AI, honestly, my, as an AI doomer, main hope is for some sort of international cooperation to slowdown and monitor the development of AI, before the Alignment problem is solved and the other risks are resolved or at least mitigated to reasonable degree. I'm probably an optimistic in this regard.

u/Lost-Basil5797 • 2 points
• 2025-05-11 20:20 UTC

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Can fucking relate, brother. I've been painting churches, lately. At least I sweat for something that will outlast me and that help people find solace. It will probably stand long after our politicians have driven us back to

the stone age.

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I'm not really worried about alignment, I'd guess this is an issue mostly related to AI takeover? Haven't given much thought to that part of the matter. I see potentials for dramatic failure of our society well before that point, namely, our historical inability to legally frame stuff that manipulate our thoughts and ways to think. It's too useful for those in power, which might explain this apparent blind spot. Still, we have so many forms of medias all trying to fuck with our brains, we let highly skilled manipulators speak almost directly to our children through ads on TVs, and now social networks...

It's all been in profit of that awful chimera of states and corporations, and we've never been to that stage before. So maybe they don't realize fully what they're doing, but I think this "fucking with people's brain" can definitely go too far, even without extreme events.

Aaaanyway. Back to
benches eh.